Requirements for CE Network Objects

2025

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1. Primary Data Structure

1.1 Import from CSV file

First row represents field name, and 2nd, 3rd, 4th, etc represents value.

	A	B	C	D	E	F
1	cell_name	latitude	longitude	height	azimuth	tilt
2	VL001EA	54.690278	25.425	32	80	0
3	VL001EB	54.690278	25.425	31	200	0
4	VL001EC	54.690278	25.425	32	320	0

Field names are not crucial, as CE includes a mapping function to align external database fields with the CE database structure.

1.2 Import from Organization's Portal

The point layer shared on the Organization's ArcGIS Enterpris Portal or ArcGIS Online can be imported into the CE Database. Field names are not crucial, as CE includes a mapping function to align external database fields with the CE database structure.

2. Cells

2.1 Minimum requirements

Value	CE Field	Units	Example	Description
Latitude	y	Degrees	49.9993	Y point coordinate in Decimal degrees and in WGS 1984 geographical coordinate system.
Longitude	x	Degrees	33.6573	X point coordinate in Decimal degrees and in WGS 1984 geographical coordinate system.
Unique identification	cell_name	[text]	5G cell XXYY	Represents cell identification, usually name.

2.2 Optional information

2.2.1 Height

Antenna (cell) height in meters above ground level. If the Mobile Operator cannot provide this information, a default value can be applied based on land use types. For example, in rural areas where towers are commonly used, the height might be set to 60 meters.

2.2.2 Azimuth

Antenna (cell) direction relative to true north, with values ranging from 0 to 359 degrees. If the Mobile Operator cannot provide the cell azimuth value, the following can be applied:

- Set the value to 0 and designate the antenna type as Omnidirectional.
- Automatically assign azimuth values based on the number of antennas on the tower. For example, if there are 3 antennas, the azimuths would be 0, 120, and 240 degrees.

2.2.3 Mechanical tilt

Antenna (cell) vertical direction.

If Mobile Operator cannot provide Cell tilt value then we can use value – 0.

2.2.4 Frequency

Cell frequency value in MHz.

If the frequency is missing, it can be determined based on the following:

- The operator's technology.
- The frequency bands associated with this technology and operator.

2.2.5 Power

Cell power in dBm.

2.2.6 Antenna gain

Antenna gain in dBi, which is taken automatically from defined antenna for Cell.

2.2.7 Losses

Miscellaneous losses in dB, which shows total losses for cables, feeders, etc.

2.2.8 Bandwidth

Cell bandwidth value in MHz. Especially required for 3G, 4G, and 5G technologies.

If Mobile Operators cannot provide this parameter, then we can apply default value based on technology and frequency band.

2.2.9 Subcarrier spacing

Especially required for 5G, while 4G uses constant value – 15. Units are in kHz.

If Mobile Operators cannot provide this parameter for 5G technology, then we can apply default value based on frequency band.

2.2.10 TX MIMO

Transmitter MIMO configuration, with possible values of 1, 2, 4, 8, 16, 32, 64.

If Mobile Operators cannot provide this parameter, a default value of 1 is typically used.

2.2.11 RX MIMO

Receiver MIMO configuration, with possible values of 1, 2, 4, 8, 16, 32, 64.

If Mobile Operators cannot provide this parameter, a default value of 1 is typically used.

2.2.12 Technology

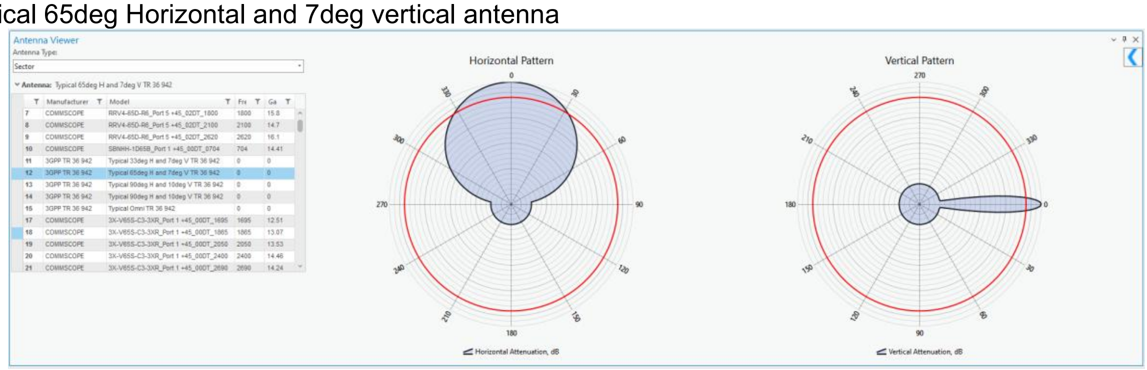
Antenna (cell) technology, with possible values including 2G, 3G, 4G, 5G, and WiFi. If the Mobile Operator cannot provide this parameter, we can:

- Estimate the technology based on the frequency.
- Default to 2G if only Signal Strength calculation is needed.

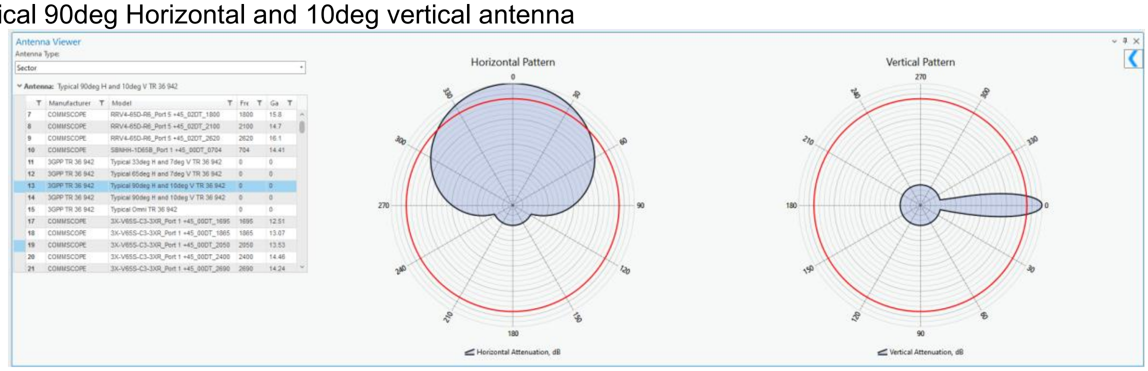
2.2.13 Antenna name

The antenna should be imported into the database, with its ID assigned to each Cell object.
If Mobile Operators cannot provide specific antenna names and patterns, a worst-case scenario can be applied using typical antennas generated according to 3GPP standards.

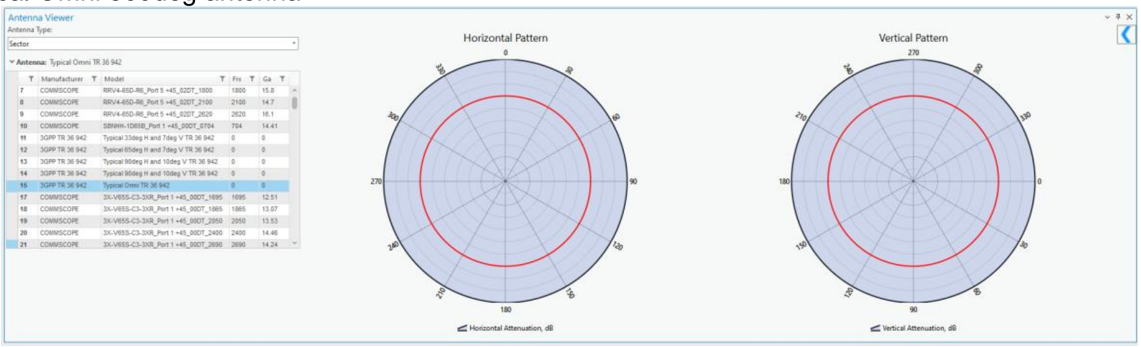
Typical 65deg Horizontal and 7deg vertical antenna



Typical 90deg Horizontal and 10deg vertical antenna



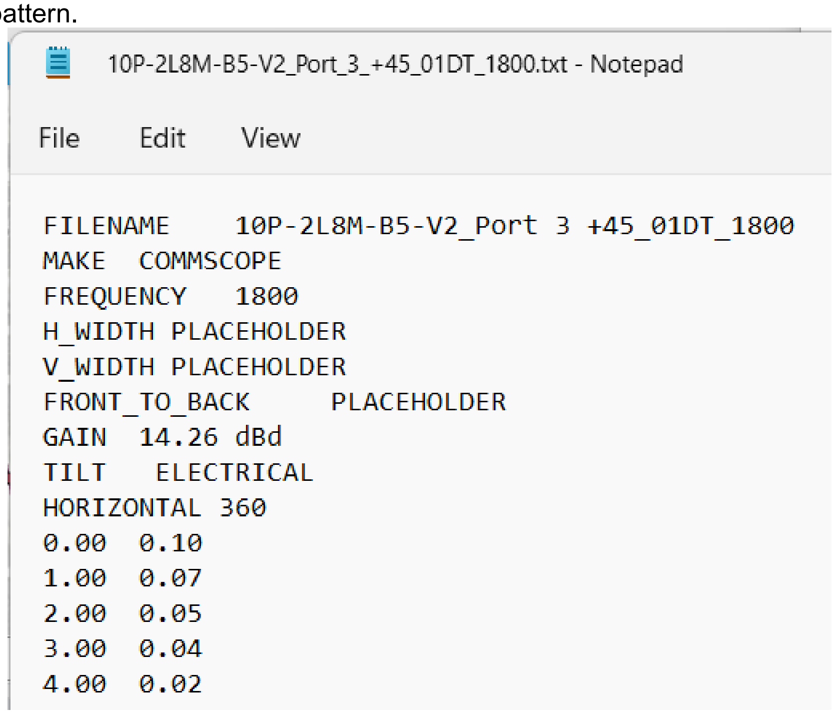
Typical Omni 360deg antenna



2.2.13.1 Antenna pattern file structure

The text file in Planet format represents:

- Main antenna parameters, such as name, gain value.
- Horizontal pattern.
- Vertical pattern.



10P-2L8M-B5-V2_Port_3_+45_01DT_1800.txt - Notepad

File Edit View

```
FILENAME    10P-2L8M-B5-V2_Port 3 +45_01DT_1800
MAKE  COMMSCOPE
FREQUENCY   1800
H_WIDTH PLACEHOLDER
V_WIDTH PLACEHOLDER
FRONT_TO_BACK     PLACEHOLDER
GAIN  14.26 dBd
TILT   ELECTRICAL
HORIZONTAL 360
0.00  0.10
1.00  0.07
2.00  0.05
3.00  0.04
4.00  0.02
```

Such files can be imported into Cellular Expert database.

2.2.14 Site (Tower) identification

Represents the Site (Tower) name in text format. This information enables the automatic creation of Site objects along with Cell objects. If the information is missing, it will be skipped.

3. Sites (optional)

Sites as an object can be imported and visualized in the project.

Value	CE Field	Units	Example	Description
Latitude	latitude	Degrees	49.9993	Y point coordinate in Decimal degrees and in WGS 1984 geographical coordinate system.
Longitude	longitude	Degrees	33.6573	X point coordinate in Decimal degrees and in WGS 1984 geographical coordinate system.
Site identification	site_name	[text]	Site 55 ID	Represents site identification, usually name.